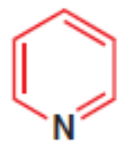
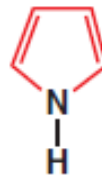


Chapter 7

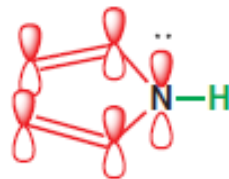
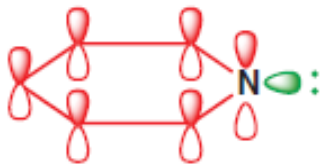
Aromatic Heterocycles



pyridine



pyrrole





Contents

7.1 Introduction

7.2 Pyridine is a very unreactive aromatic imine

7.3 Five-membered aromatic heterocycles are good at electrophilic substitution

7.4 Furan and thiophene are oxygen and sulfur analogues of pyrrole

7.5 Benzo-fused heterocycles

7.6 Fusing rings to pyridines: quinolines and isoquinolines

7.7 Synthesis of aromatic heterocycles

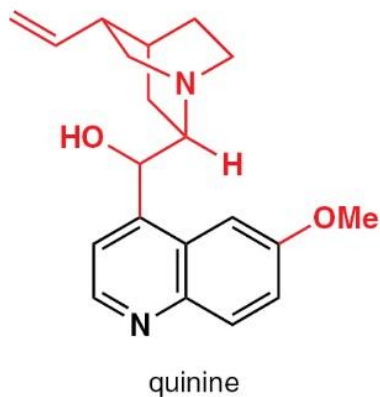
7.8 Summary

References: Chapters 29 and 30 of Clayden's Organic Chemistry

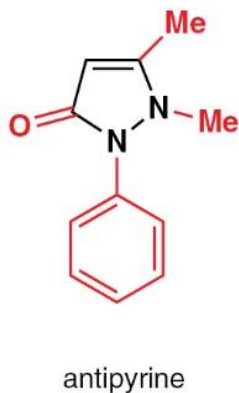


7.1 Introduction

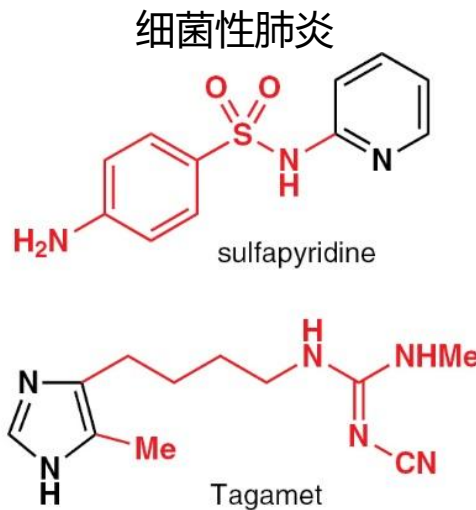
Probably about **two-thirds of** organic compounds belong to this class, and they number among them some of the **most significant compounds for human beings**. If we think only of drugs we can define the history of medicine by heterocycles.



抗疟



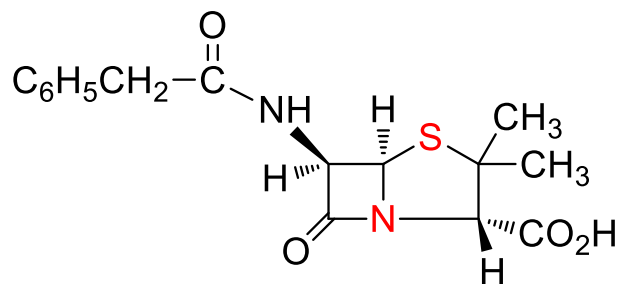
解热镇痛
(已淘汰)



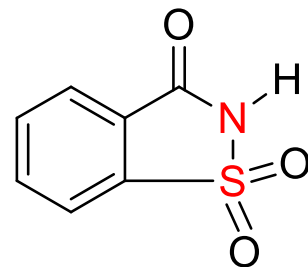
胃溃疡、肠溃疡



性功能障碍

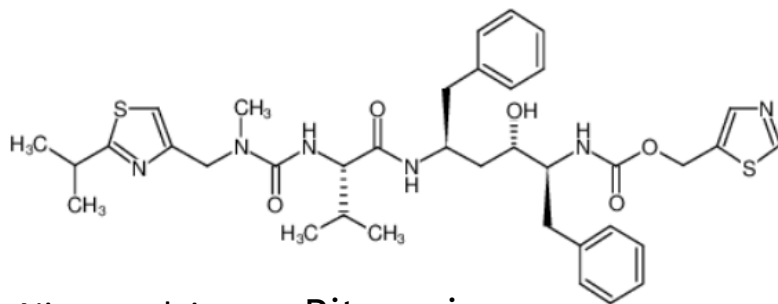
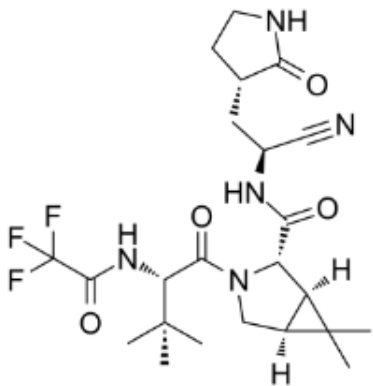
Nc1nc(=O)n([C@H]2O[C@H](CO)[C@@H](O)[C@H]2F)c1.[Cl-]

阿兹夫定 (Azvudine) **抗新冠**



Saccharin

糖精



Nirmatrelvir + Ritonavir
奈玛特韦 利托那韦

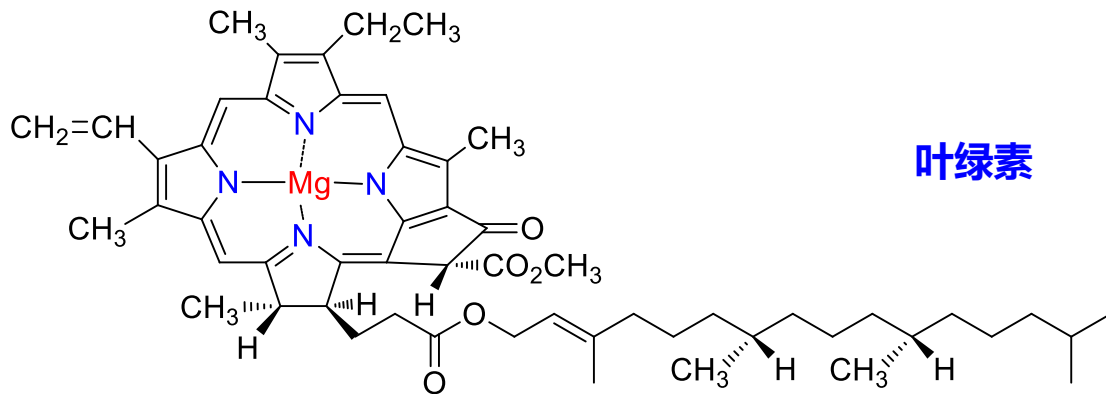
Paxlovid

**帕罗韦德
抗新冠**





自然界中的重要杂环化合物

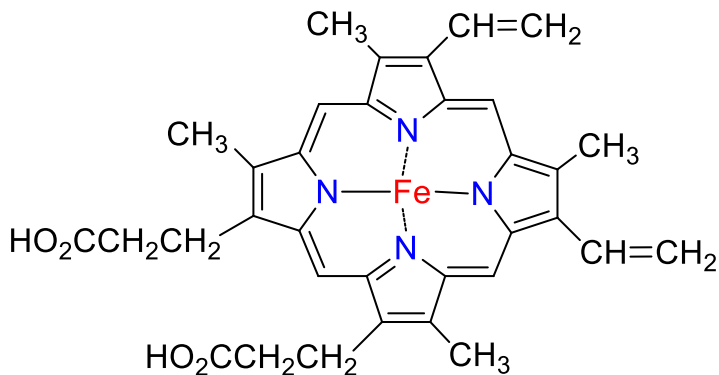


叶绿素



Porphyrin
卟啉环

Chlorophyll a



血红素



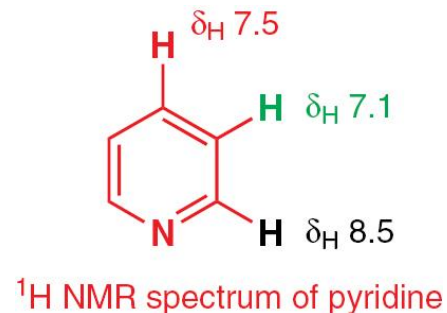
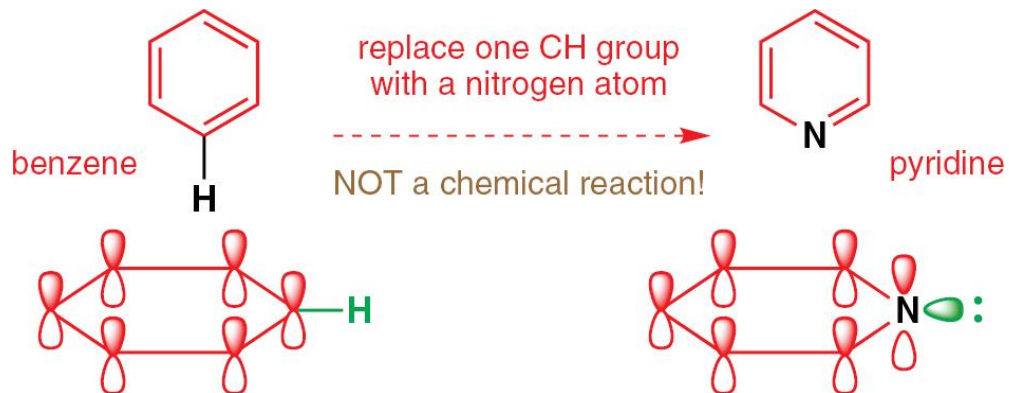
Heme

一个血红蛋白分子含有四个血红素

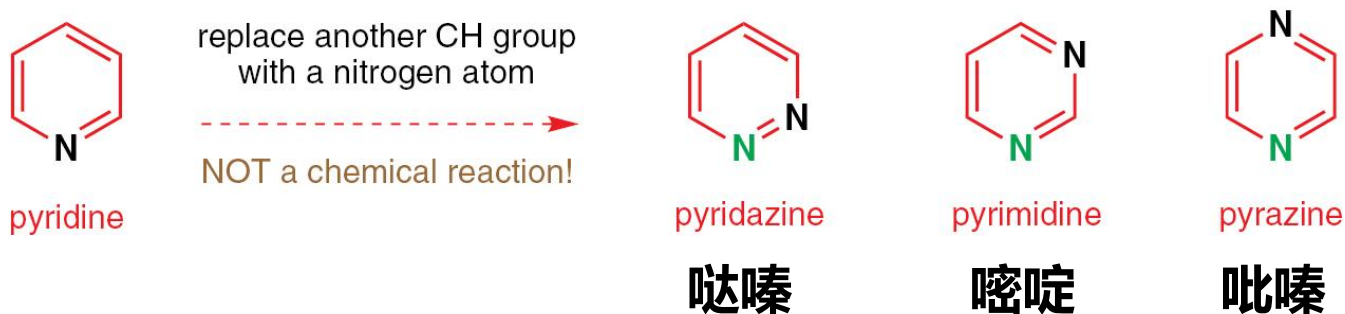


Six-membered aromatic heterocycles

Pyridine 吡啶



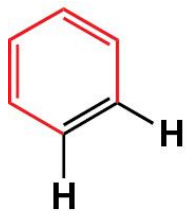
Other six-membered aromatic heterocycles





Five-membered aromatic heterocycles

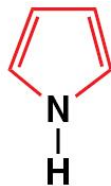
Pyrrole 吡咯



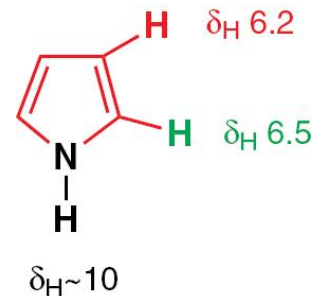
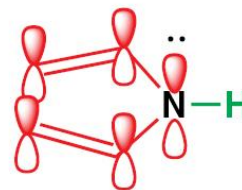
benzene

replace a CH=CH unit
with a nitrogen atom

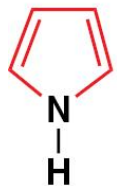
NOT a chemical reaction!



pyrrole

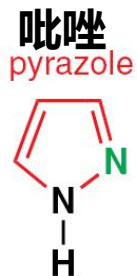


Other five-membered aromatic heterocycles

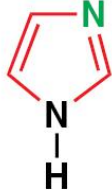


replace one
CH group
with a
nitrogen atom

NOT a
chemical
reaction!



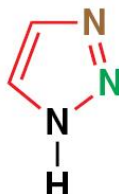
吡唑
pyrazole



咪唑
imidazole

replace a second
CH group
with a
nitrogen atom

NOT a
chemical
reaction!



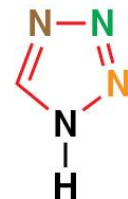
1,2,3-三氮唑
1,2,3-triazole



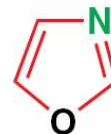
1,2,4-三氮唑
1,2,4-triazole

replace a third
CH group
with a
nitrogen atom

NOT a
chemical
reaction!



四氮唑
tetrazole



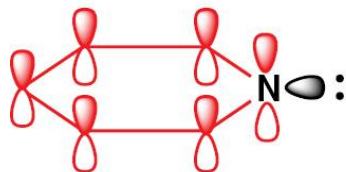
噁唑
oxazole



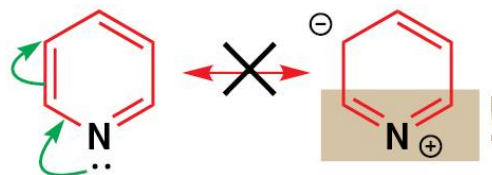
噻唑
thiazole



7.2 Pyridine is a very unreactive aromatic imine

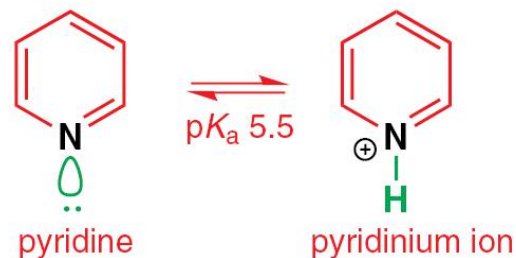
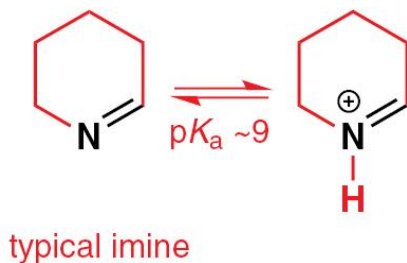
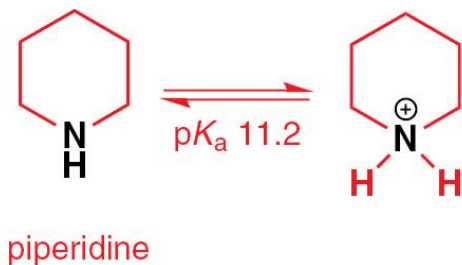


lone pair in sp^2 orbital at right angles to p orbitals in ring:
no interaction between orthogonal orbitals



attempts to delocalize lone pair lead to absurd structures

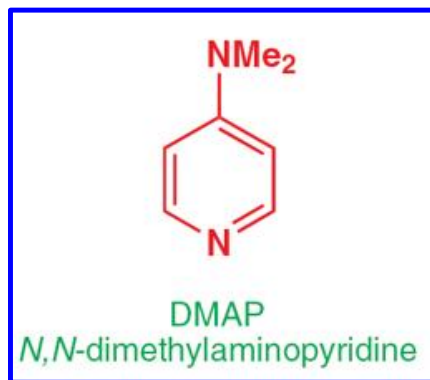
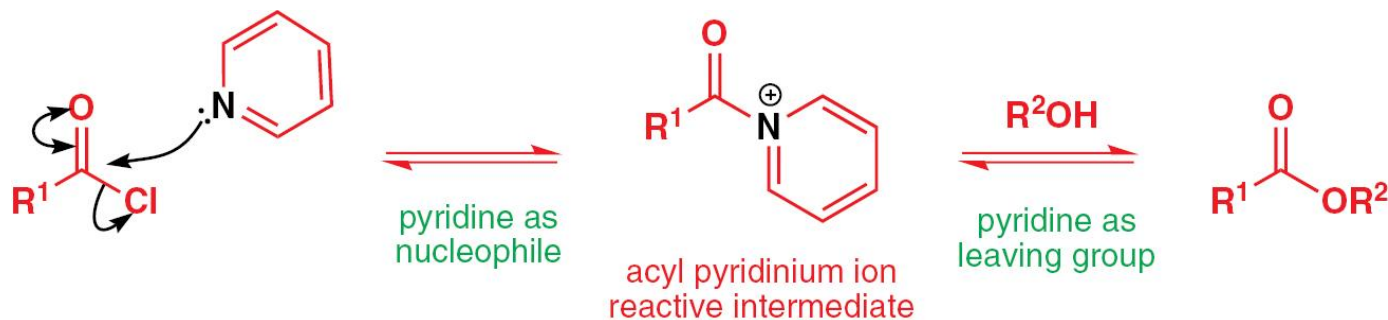
- Pyridine is a weak base with a pK_a (for its conjugate acid) of 5.5.



碱性减弱



- Pyridine is a reasonable nucleophile for carbonyl groups



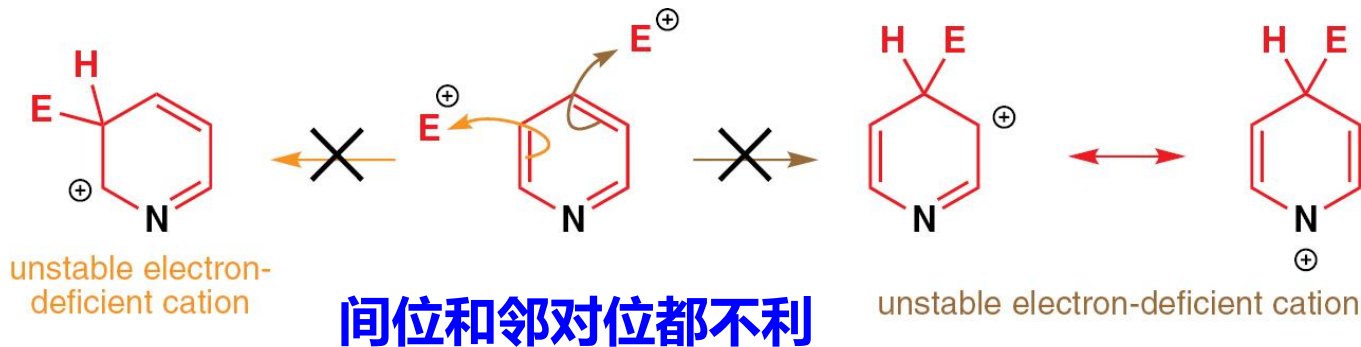
is better, and why?

常作为酯化等反应的催化剂

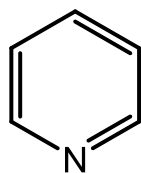


Pyridine is **bad** at electrophilic aromatic substitution

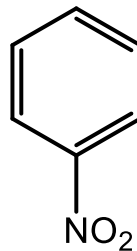
吡啶不易发生亲电取代反应



芳环上的反应性:

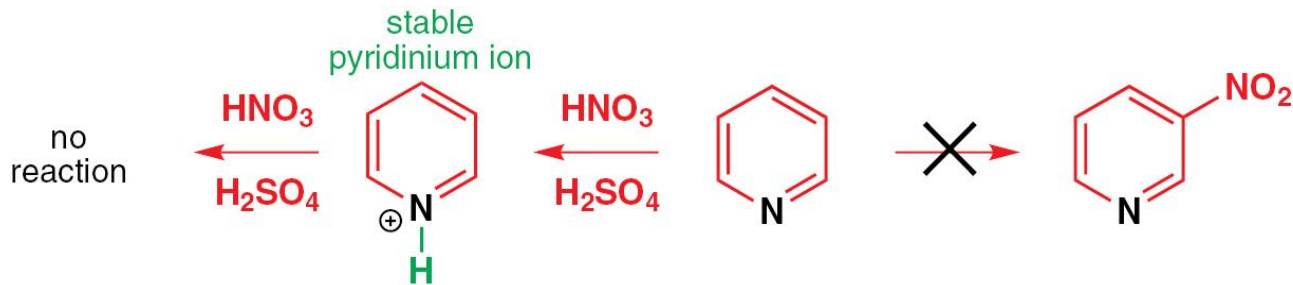


≈

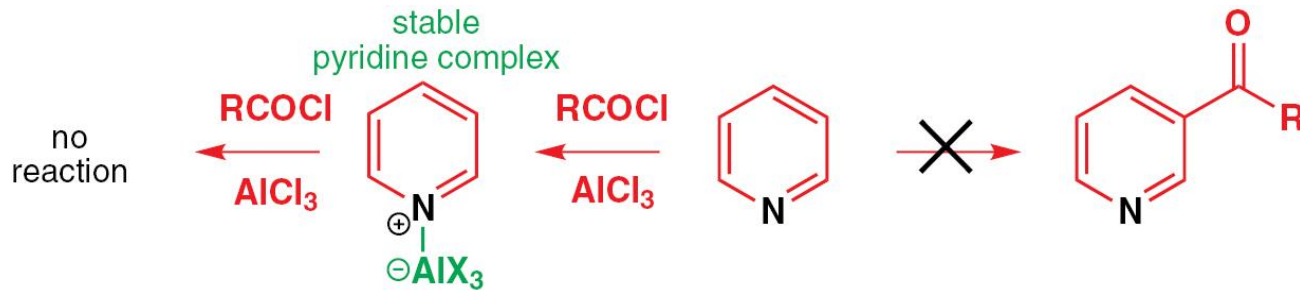




吡啶不易发生亲电取代反应



进一步降低芳环活性

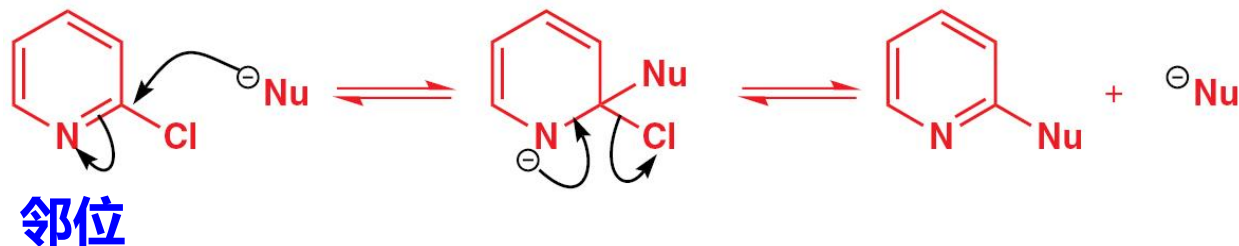


进一步降低芳环活性

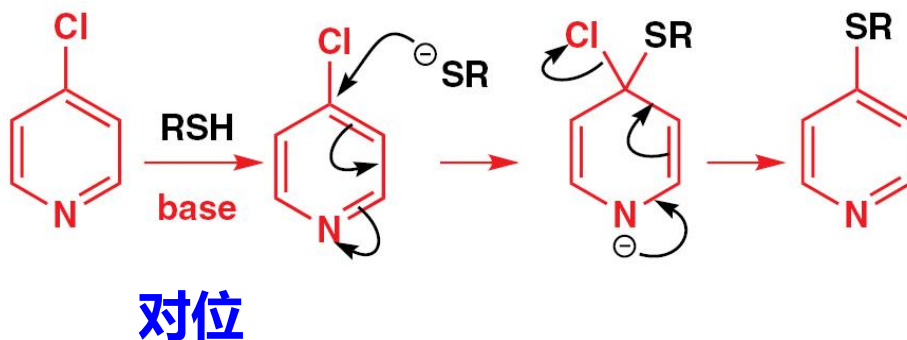
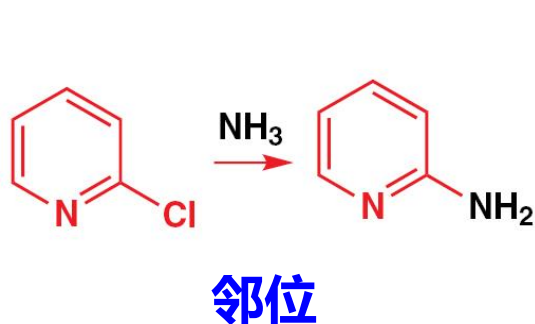


Nucleophilic substitution is easy with pyridines

邻对位卤代的吡啶可发生芳香亲核取代反应



芳香亲核取代反应



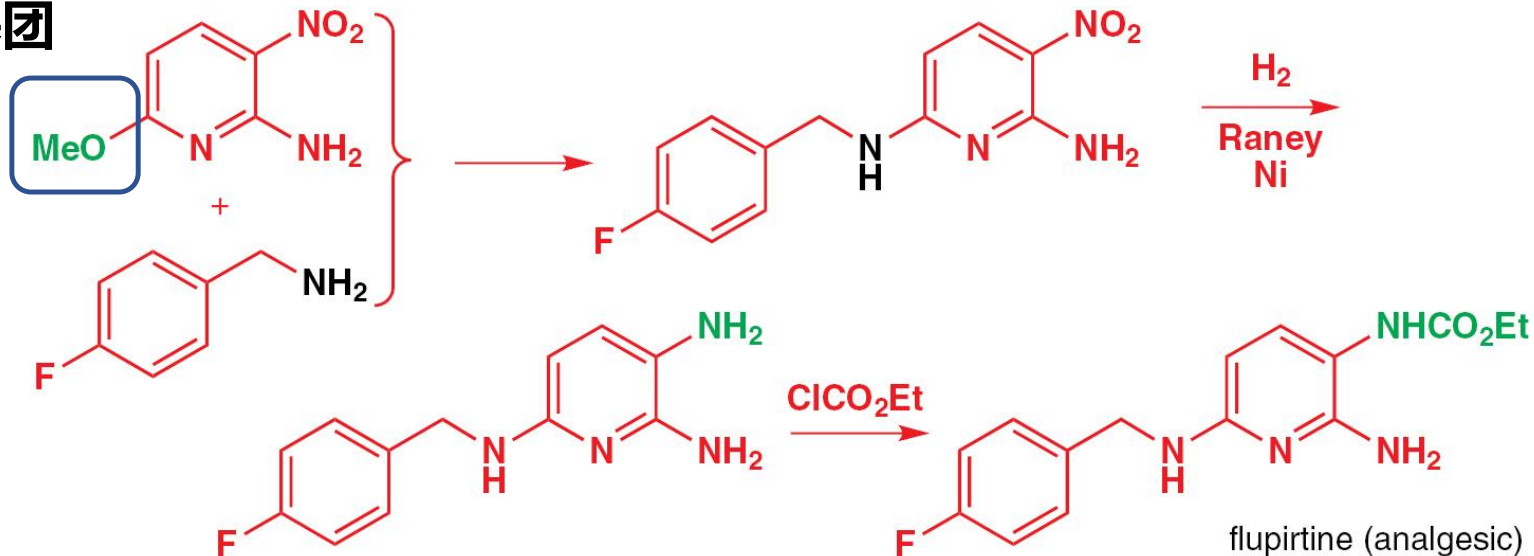


吡啶芳香亲核取代反应的应用：药物Flupirtine的合成

合适的底物设计

离去基团

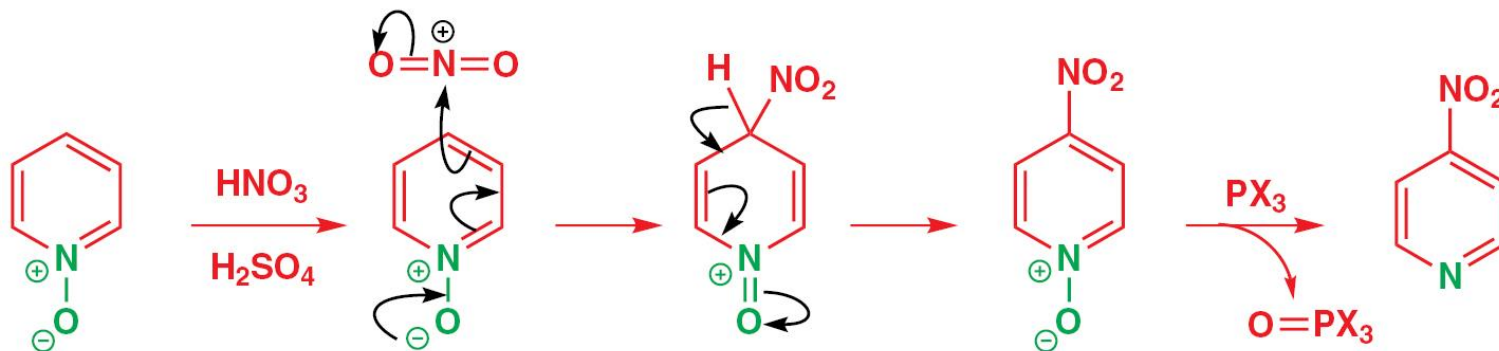
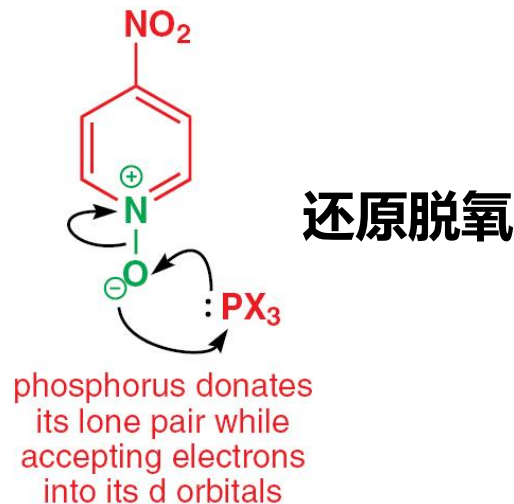
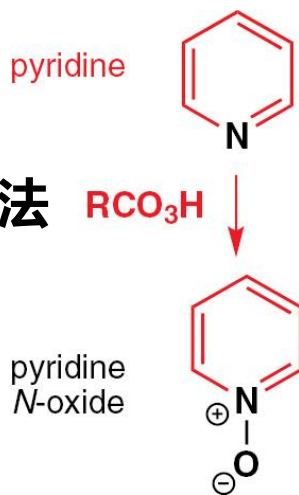
NO_2 对位
邻位





Pyridine N-oxides are reactive towards both electrophilic and nucleophilic substitution

活化吡啶的方法



吡啶氮氧化物改变亲电取代反应位点及活性

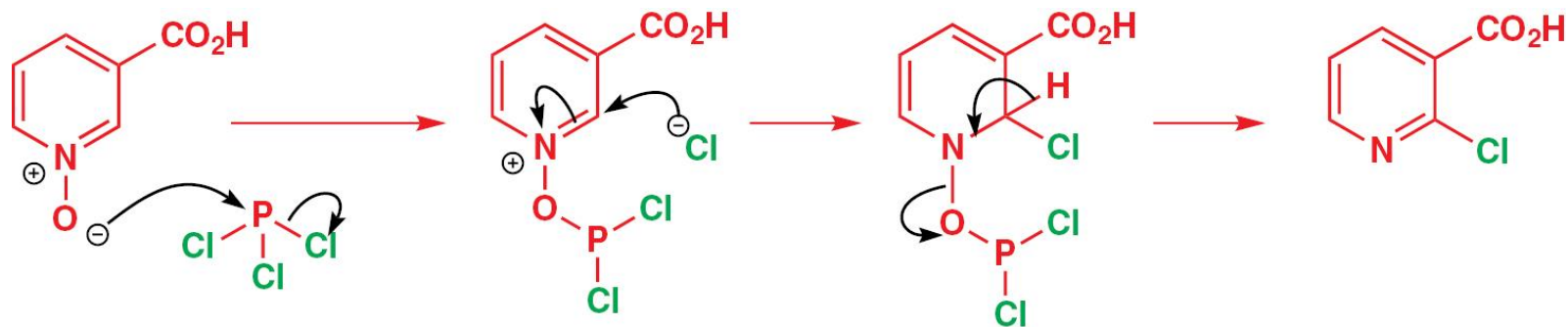
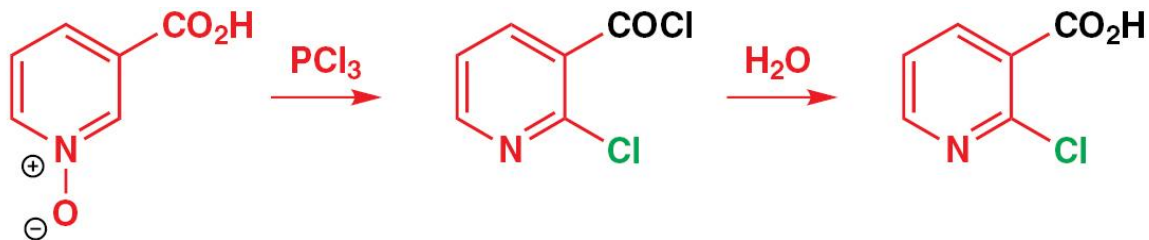


Nucleophilic substitution

活化



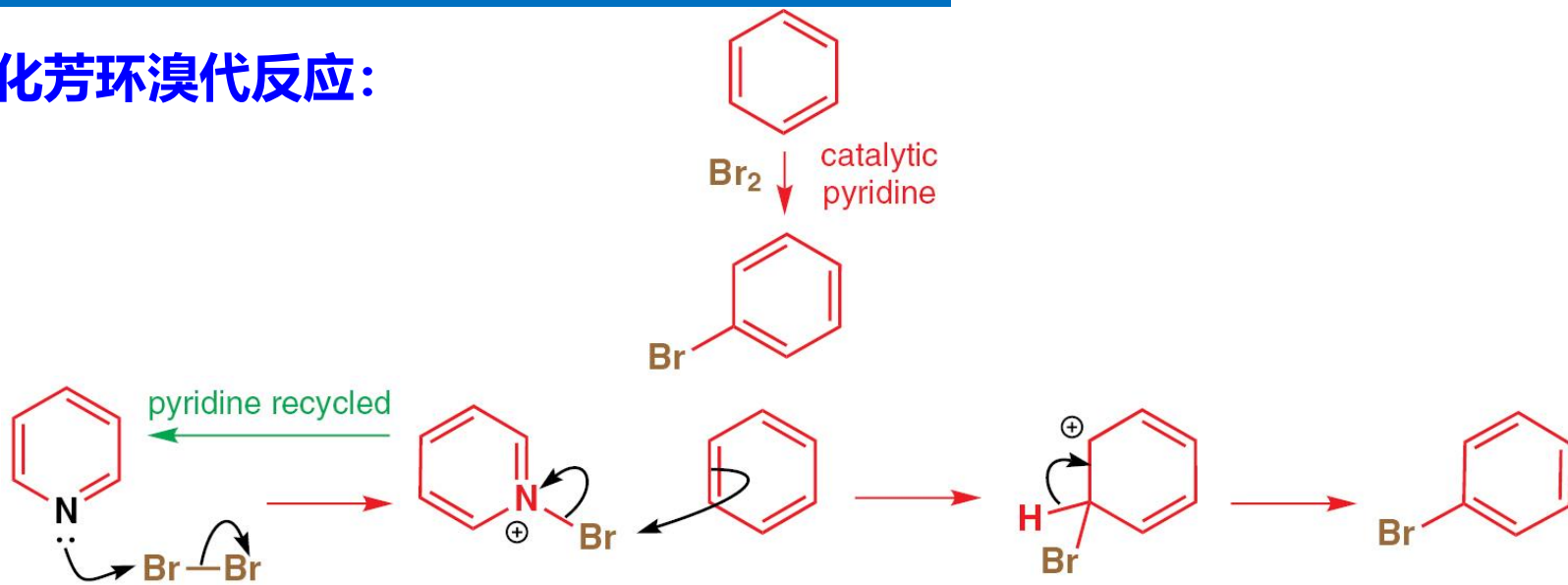
脱除



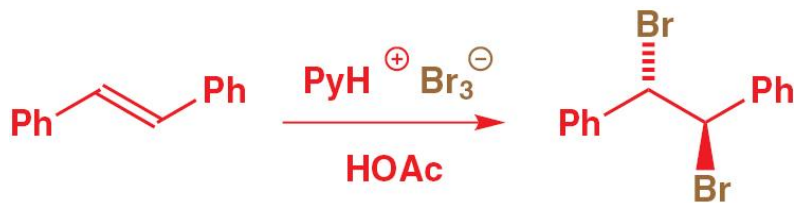
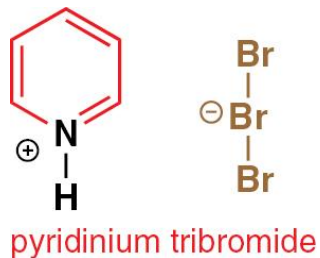


Pyridine as a catalyst and reagent

活化芳环溴代反应:



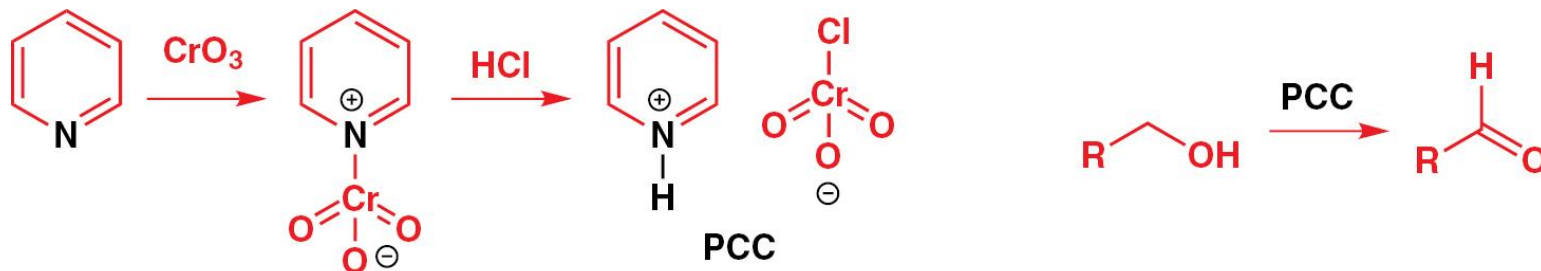
活化烯烃溴加成反应:



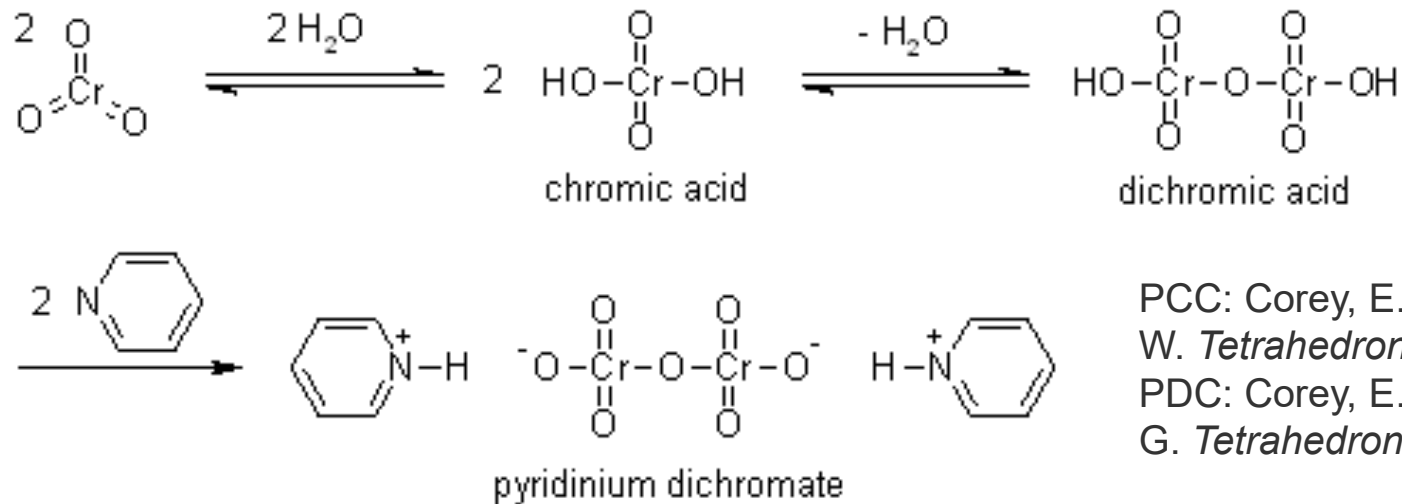


PCC与PDC氧化 (E. J. Corey)

PCC (Pyridinium Chlorochromate) 氧化:



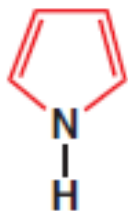
PDC (Pyridinium Dichromate) 氧化:



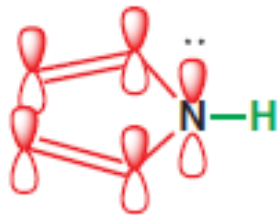
PCC: Corey, E. J.; Suggs, J. W. *Tetrahedron Lett.* **1975**, 16, 2647.
PDC: Corey, E. J.; Schmidt, G. *Tetrahedron Lett.* **1979**, 20, 399.



7.3 Five-membered aromatic heterocycles are good at electrophilic substitution



pyrrole



The HOMO should go up in energy and the ring become more nucleophilic.

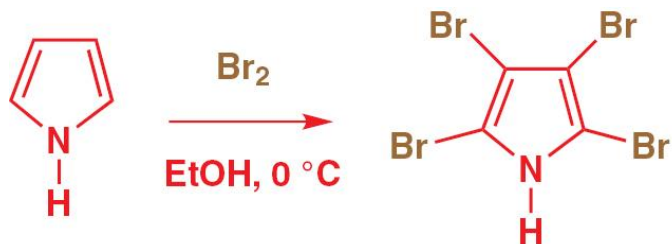
- 具有芳香性
- 比苯环更富电子
- 极弱碱



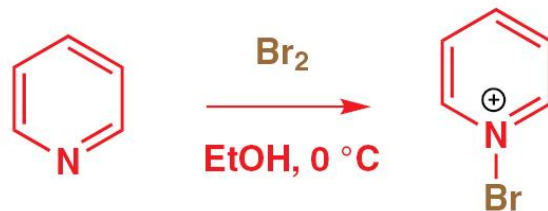
The pKa of pyrrole acting as a base is about -4 , and protonation occurs at carbon below pH -4 . By contrast, the **NH proton (pKa 16.5)** can be removed by much weaker bases than those that can remove protons on normal secondary amines.



吡咯具有很好的芳香亲电取代反应活性:

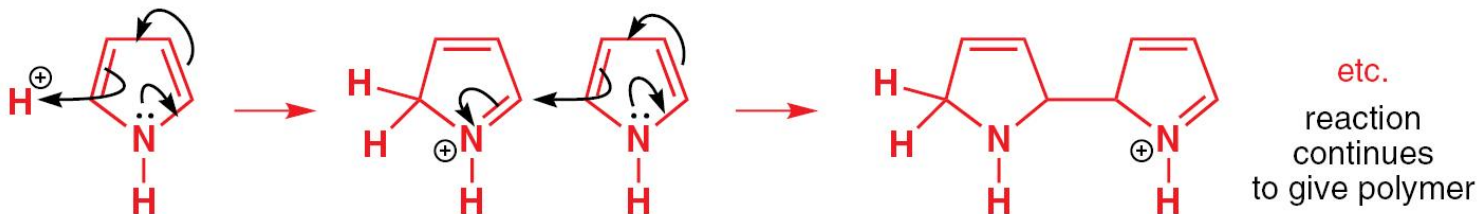


V.S.



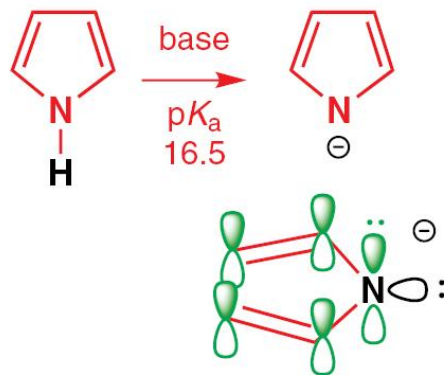
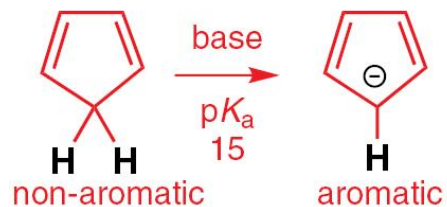
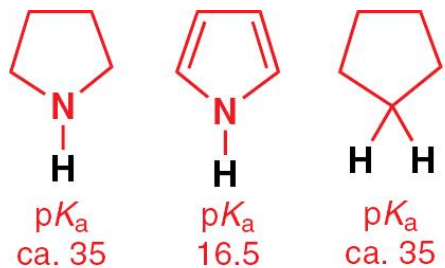
Pyrrole polymerizes! 强酸条件下发生聚合

Strong acids, those such as H_2SO_4 with a pK_a of less than -4 , cannot be used without polymerization of pyrrole.

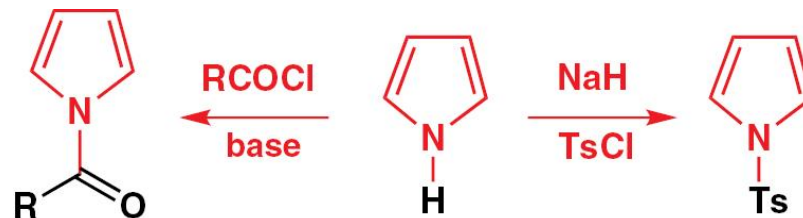




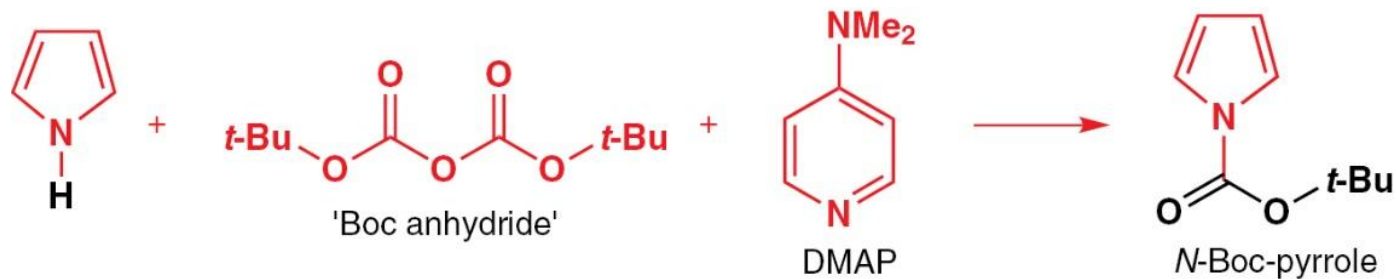
吡咯N上的反应



N-酰化、磺酰化反应:



N-酰化反应:

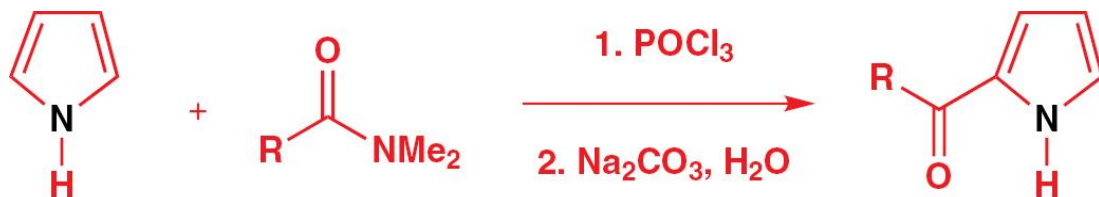
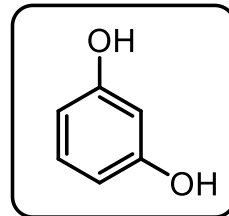




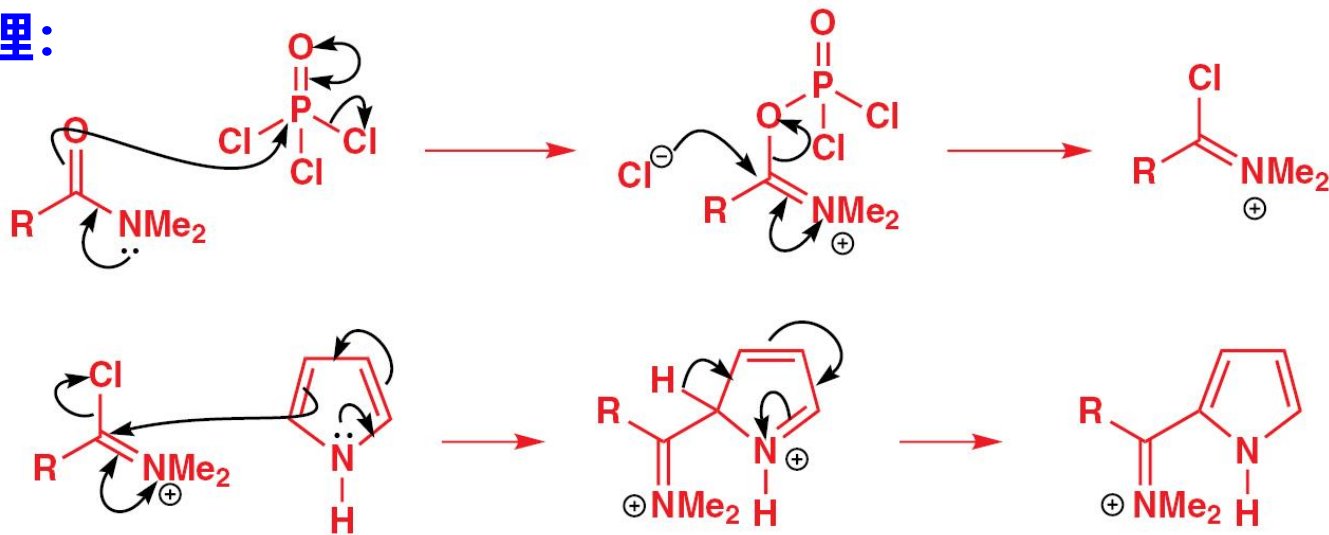
Vilsmeier reaction

通常只有富电子芳环才可发生威斯迈尔反应

位置选择性?



反应机理:

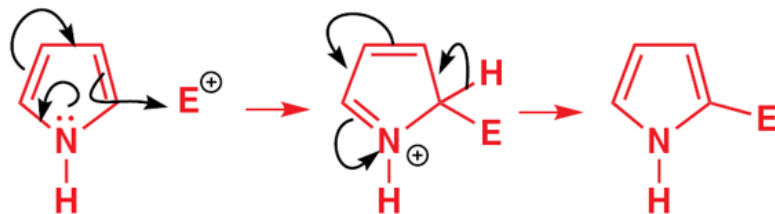




Regioselectivity of pyrrole electrophilic substitutions

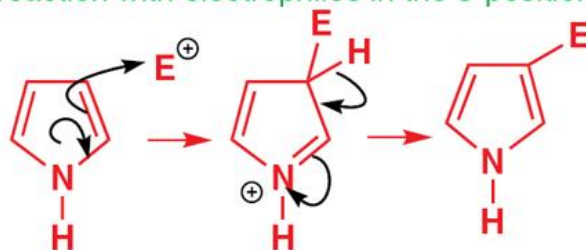
吡咯的亲电取代反应通常发生在2,5-位:

reaction with electrophiles in the 2-position



more stable

reaction with electrophiles in the 3-position

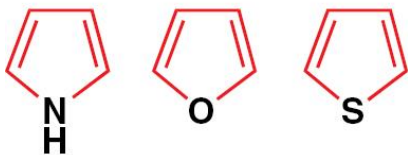


less stable

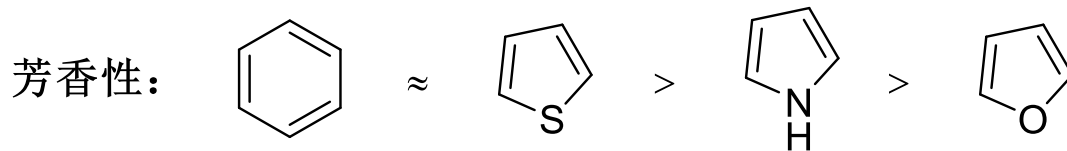


7.4 Furan and thiophene: analogues of pyrrole

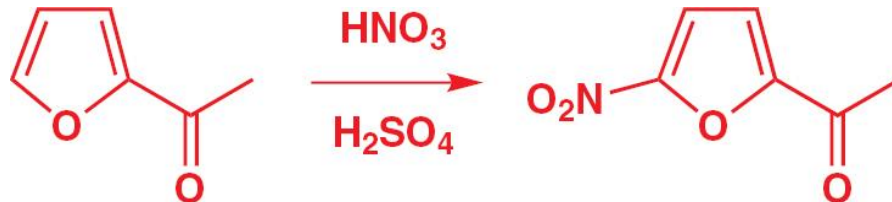
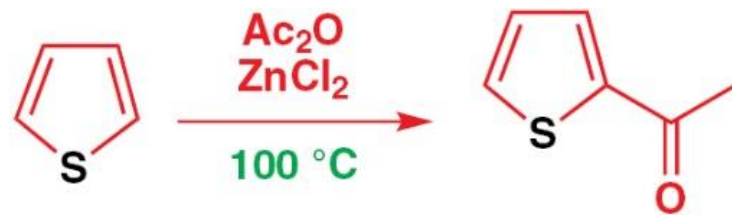
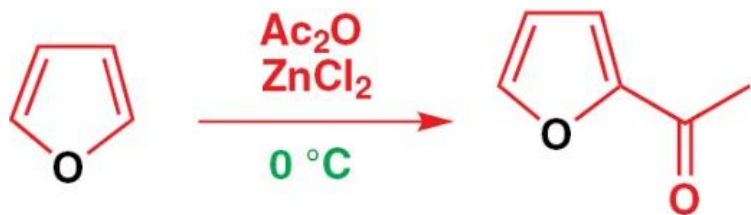
pyrrole furan thiophene



芳香性:



付克酰基化:

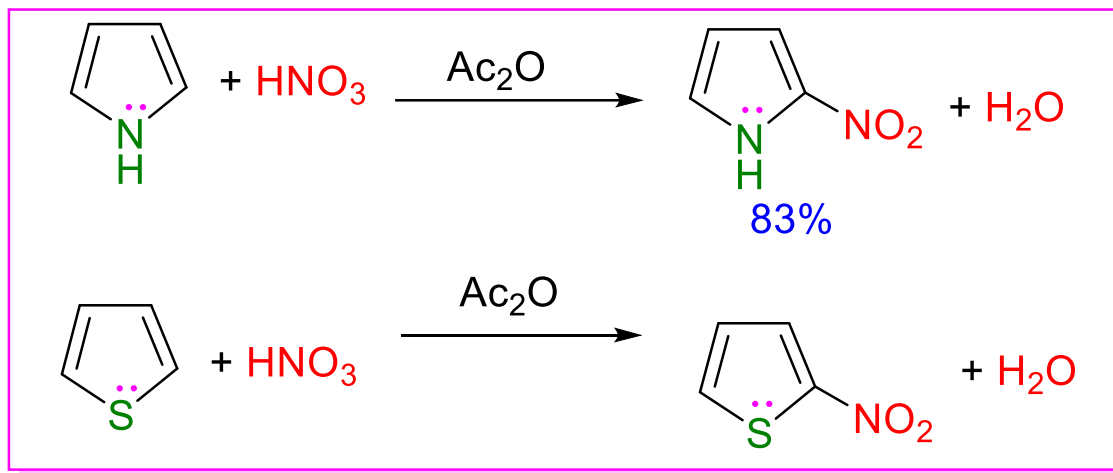




Nitration:



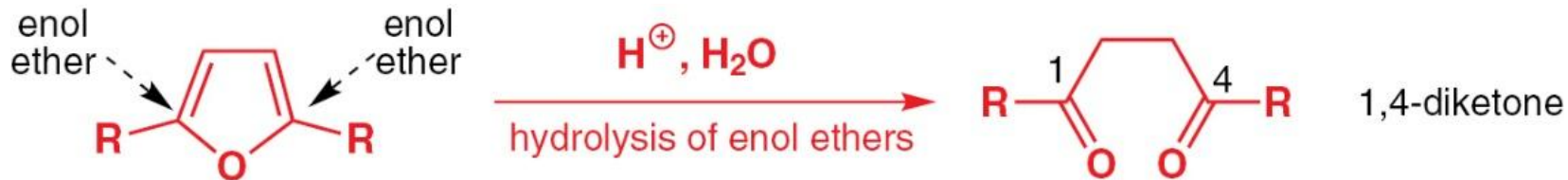
温和的硝化试剂





Reactions of furan

水解反应:



溴代反应:

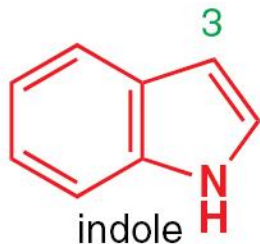
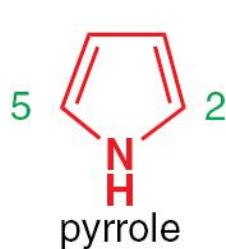


过度溴代





7.5 Benzo-fused heterocycles

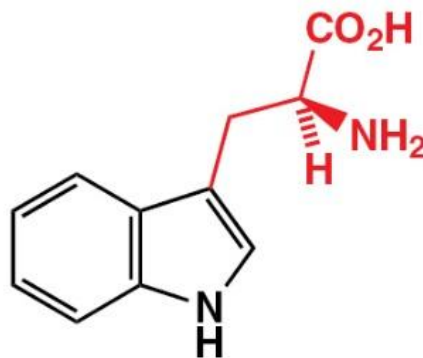
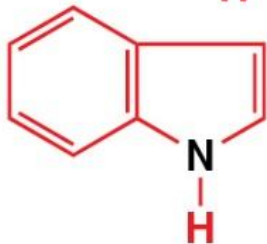


吲哚

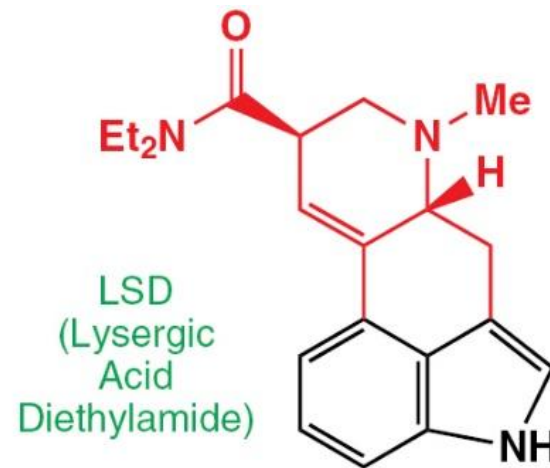
反应特点大不相同



often drawn like this:



色氨酸

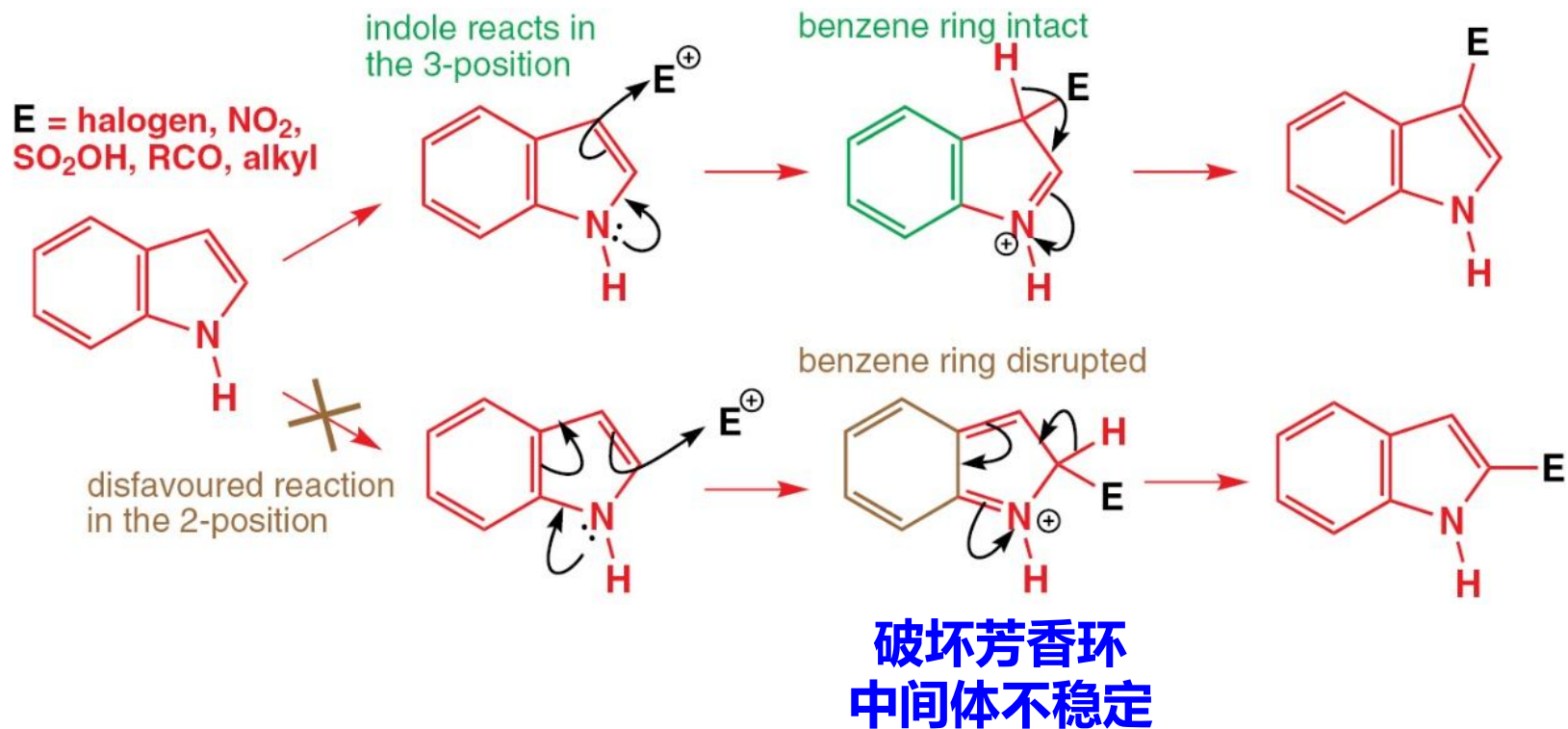


麦角酸酐二乙胺
(致幻剂)



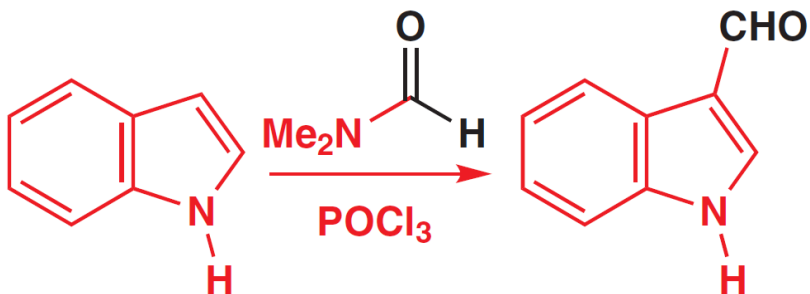
Electrophilic substitution on indole: C3 vs C2

通常发生在C3位

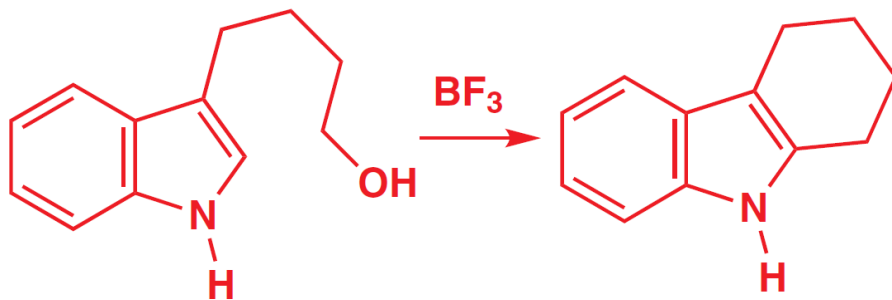




Electrophilic substitution on indole: C3 vs C2



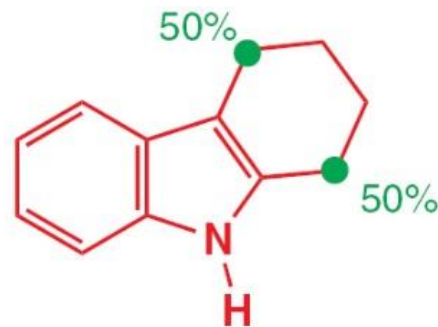
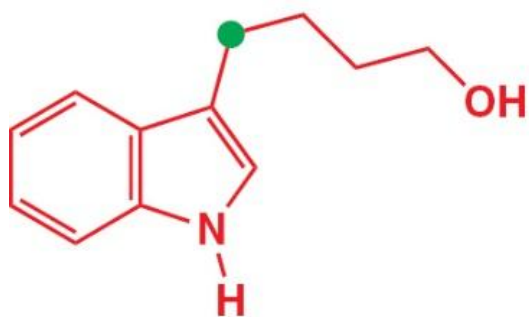
Vilsmeier reaction



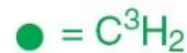
当吲哚3-位被占据后，芳香
亲电取代反应在C2位发生！？



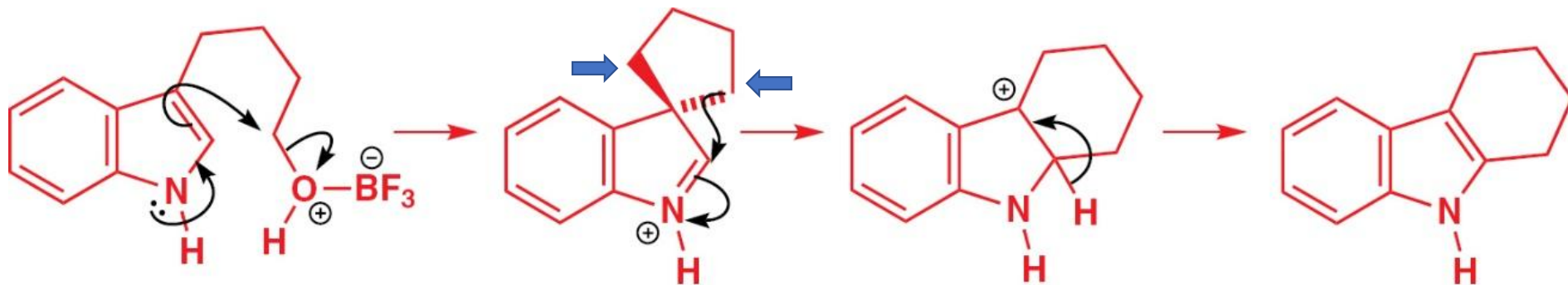
机理实验:



H同位素标记



二者均可迁移



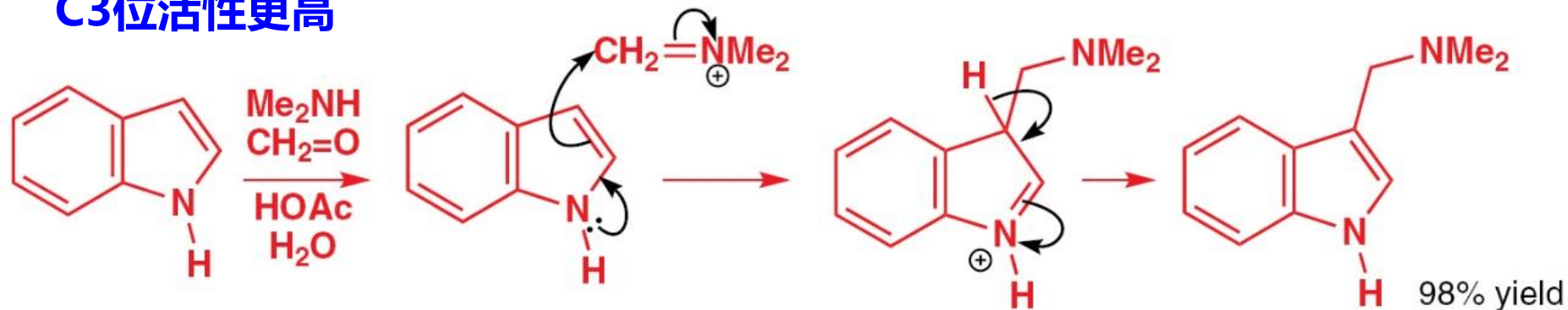
先发生在C3位

芳构化驱动重排

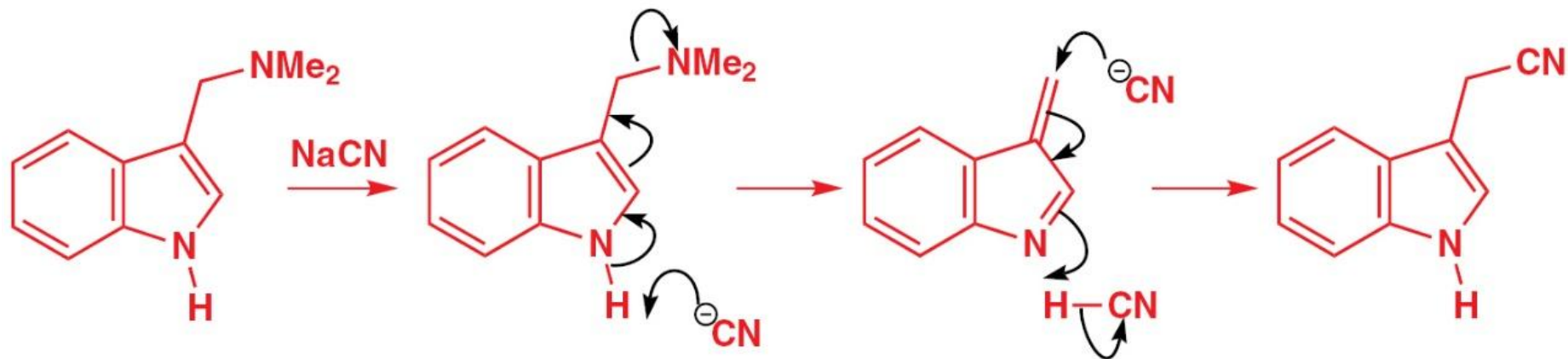


Mannich reaction on indole: C3 vs C2

C3位活性更高



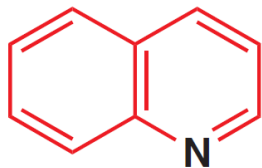
吲哚甲位继续发生取代反应：机制不同



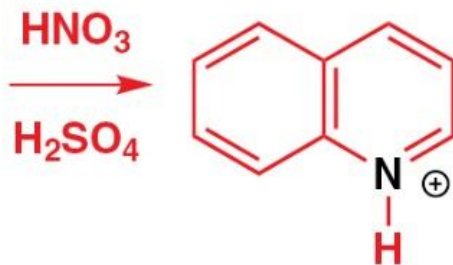
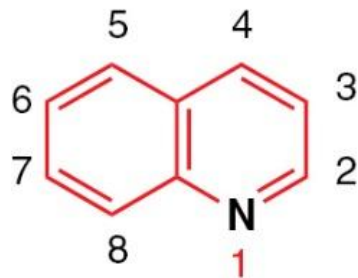


7-6 Fusing rings to pyridines: quinolines and isoquinolines

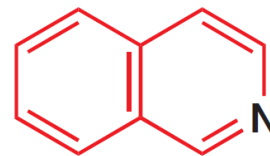
喹啉



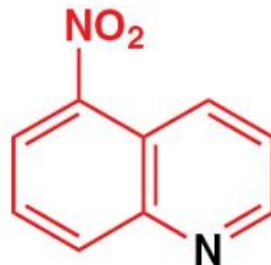
quinoline



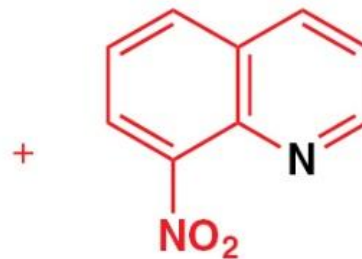
异喹啉



isoquinoline



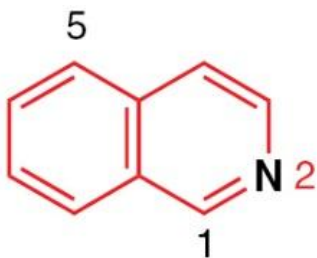
50% yield



42% yield

在苯环侧发生芳香亲电取代反应

isoquinoline



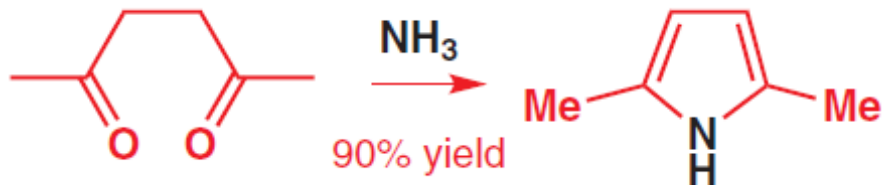
72% yield

在苯环侧发生芳香亲电取代反应

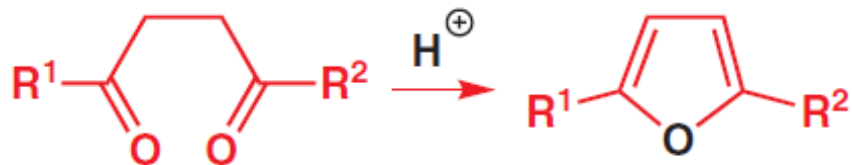


7.7 Synthesis of aromatic heterocycles

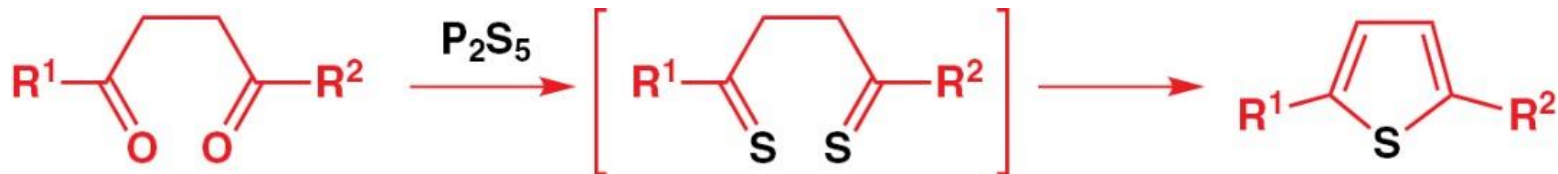
吡咯的合成:



呋喃的合成:



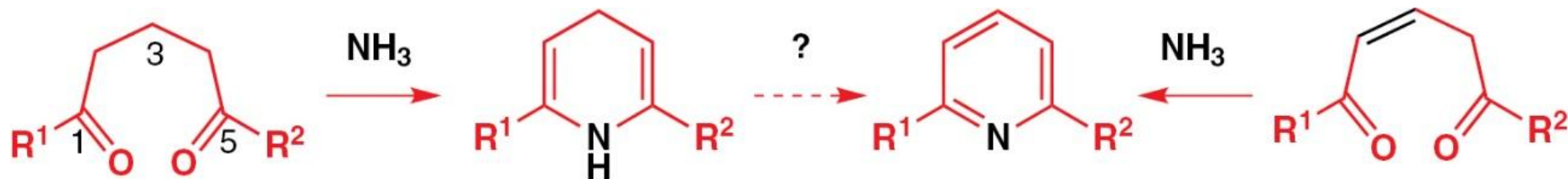
噻吩的合成:



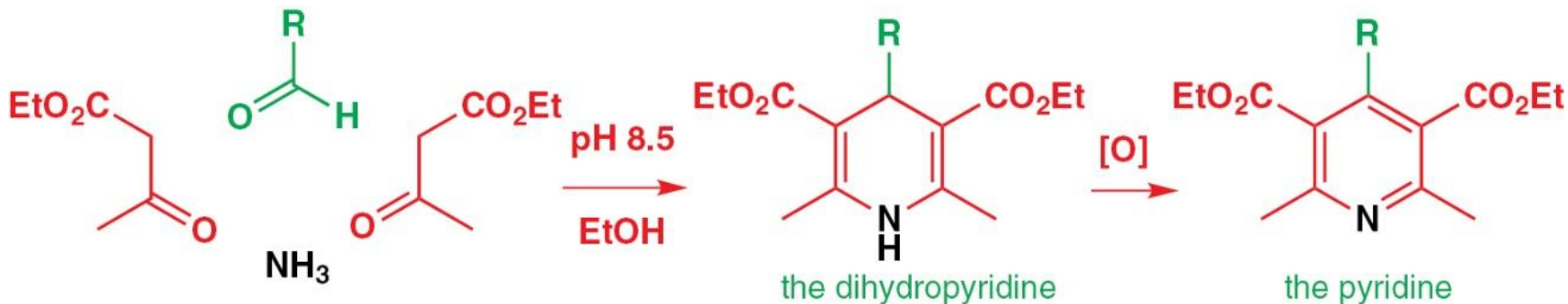


Synthesis of pyridine

吡啶合成策略:



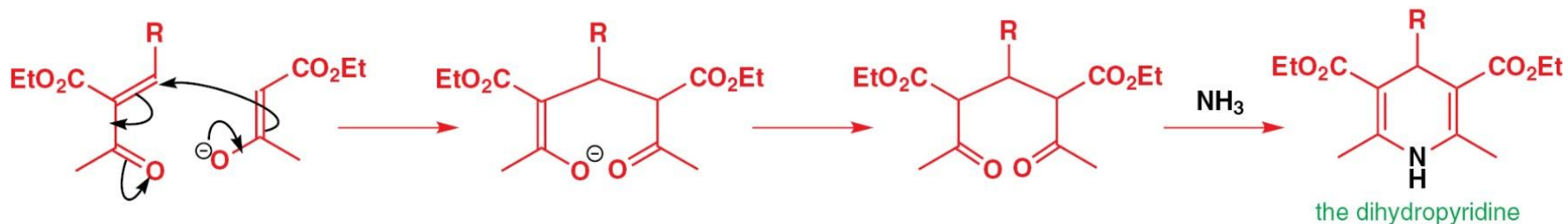
方法一：Hantzsch合成法



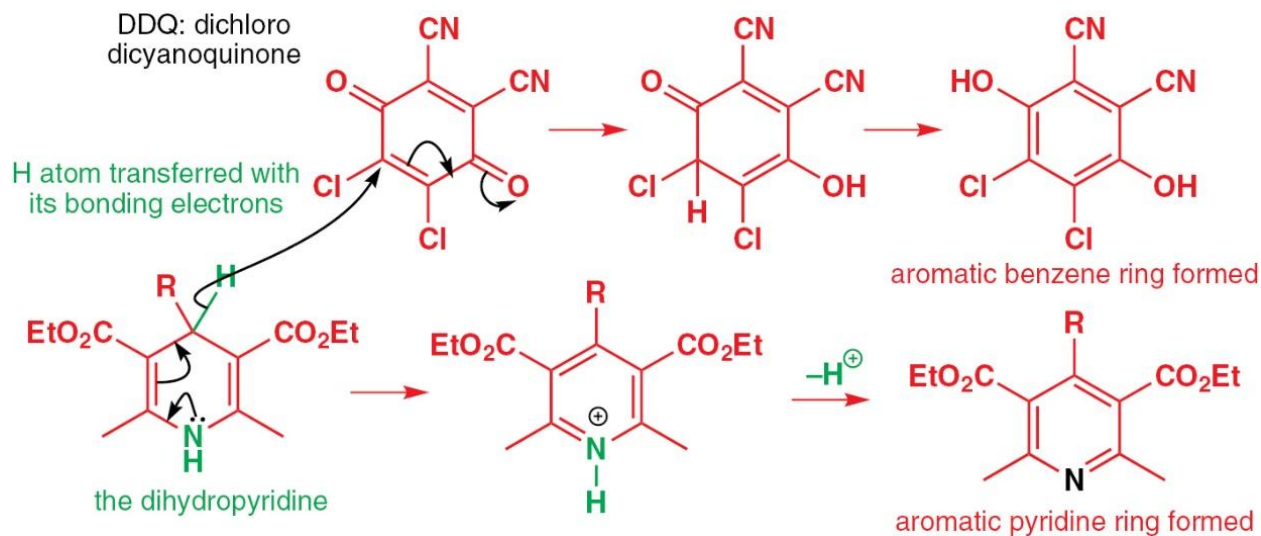


Mechanism of Hantzsch synthesis (1)

Cyclization

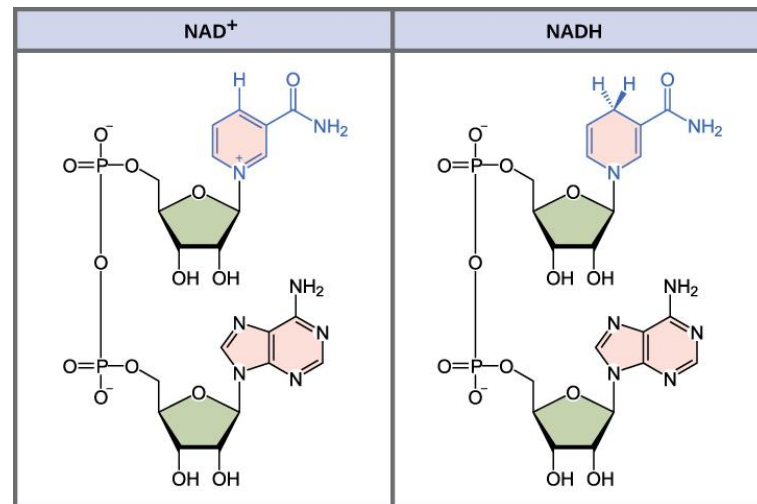
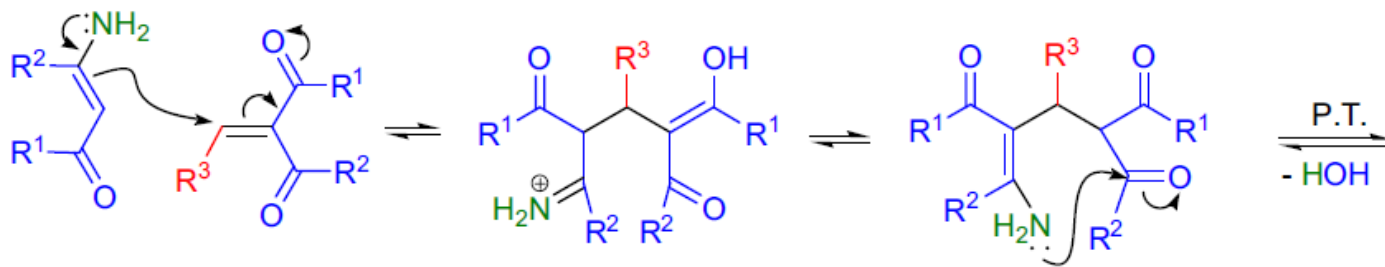
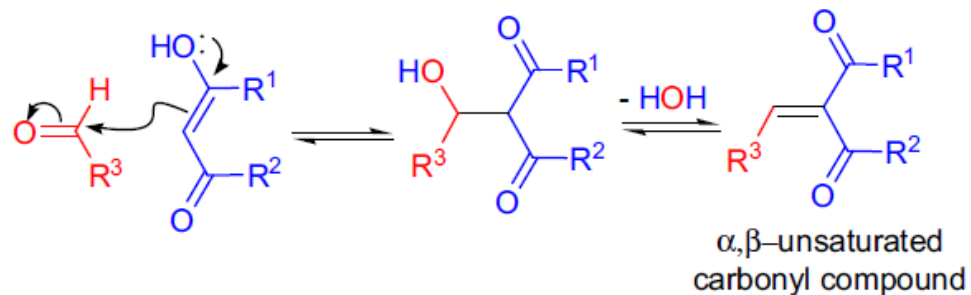
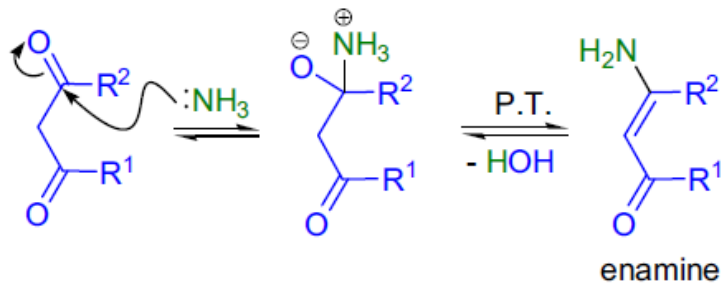


Oxidation



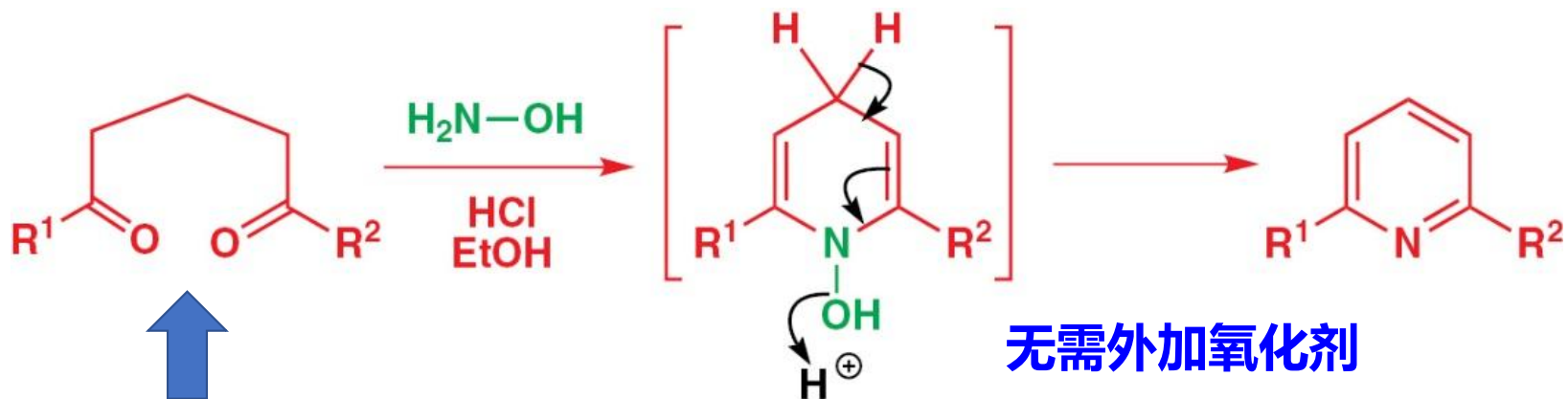


Mechanism of Hantzsch synthesis (2)

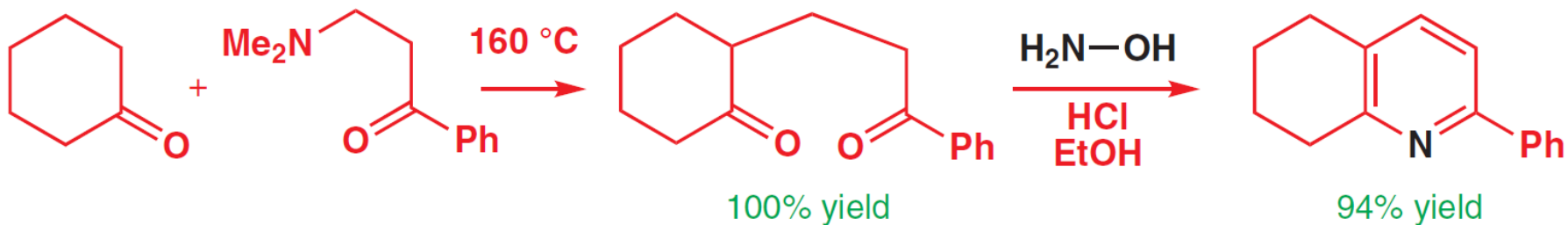




方法二：二酮（醛）羟胺缩合法

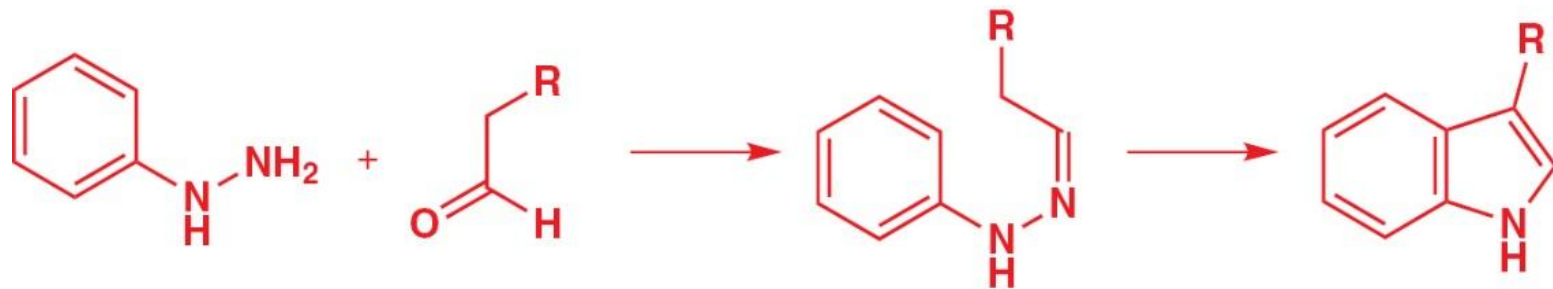
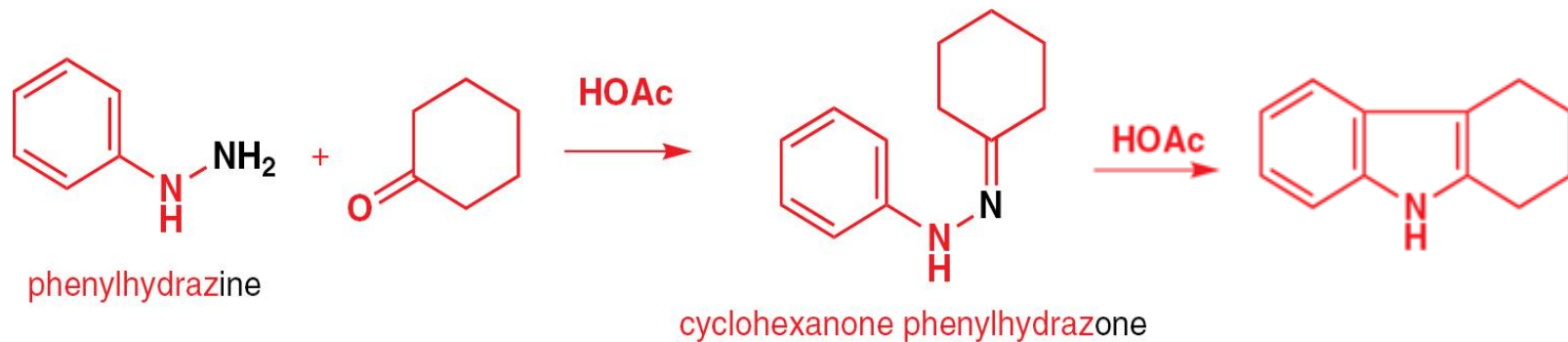


如何合成?





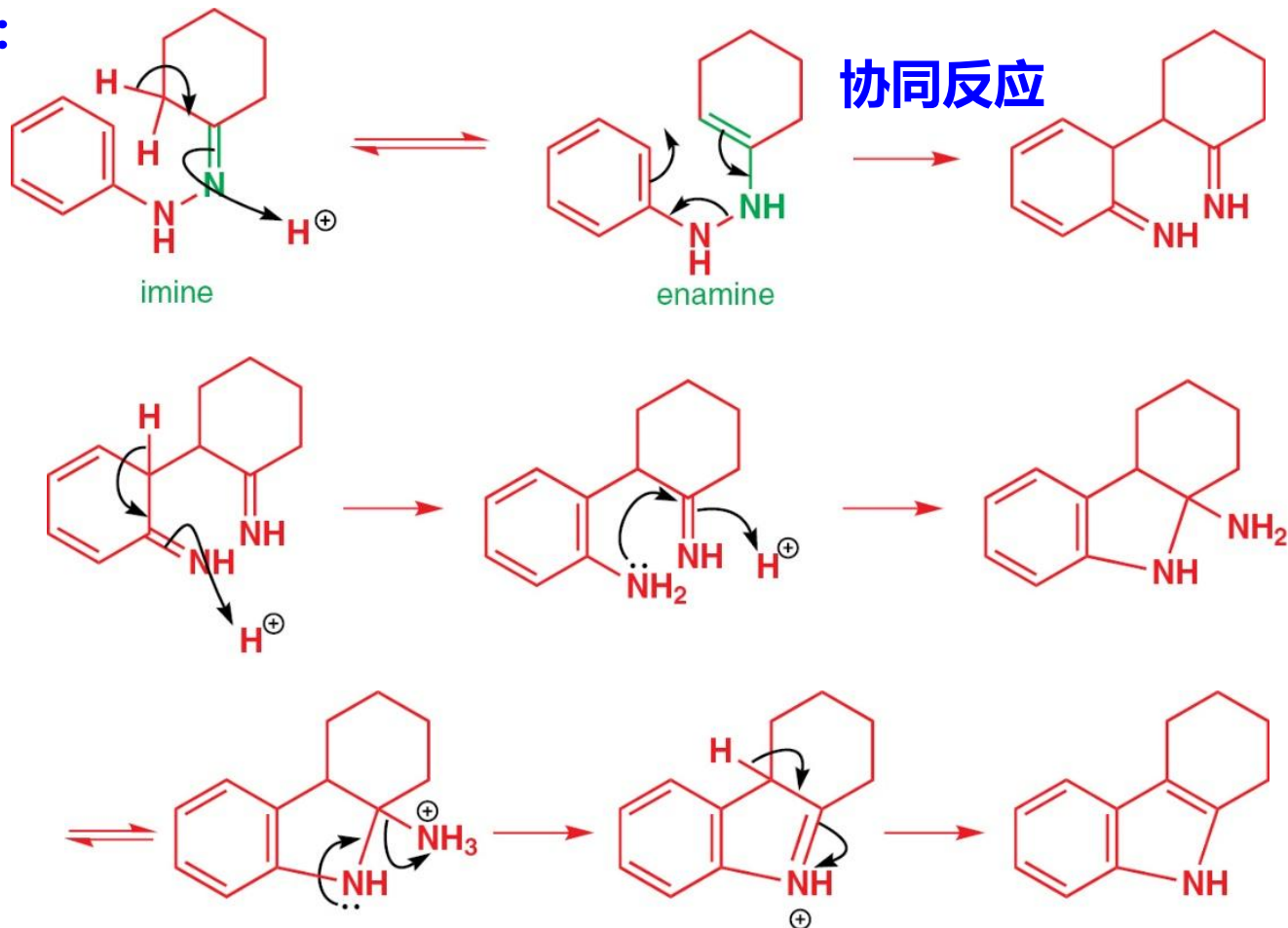
Fisher indole synthesis

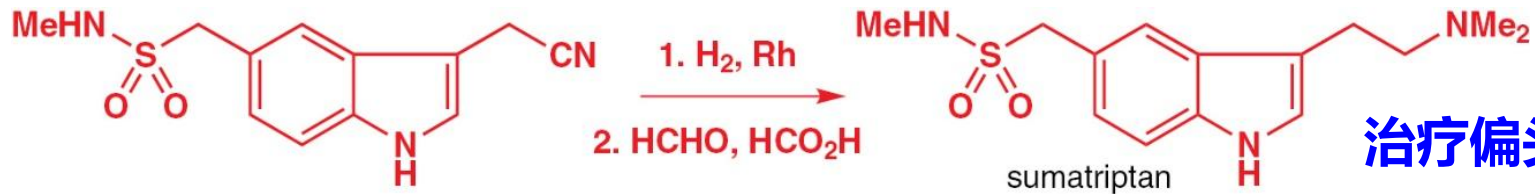
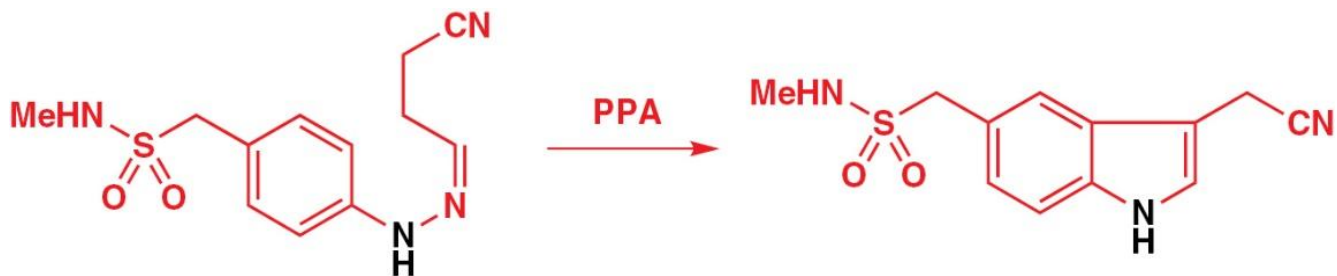




Fisher indole synthesis

反应机理:



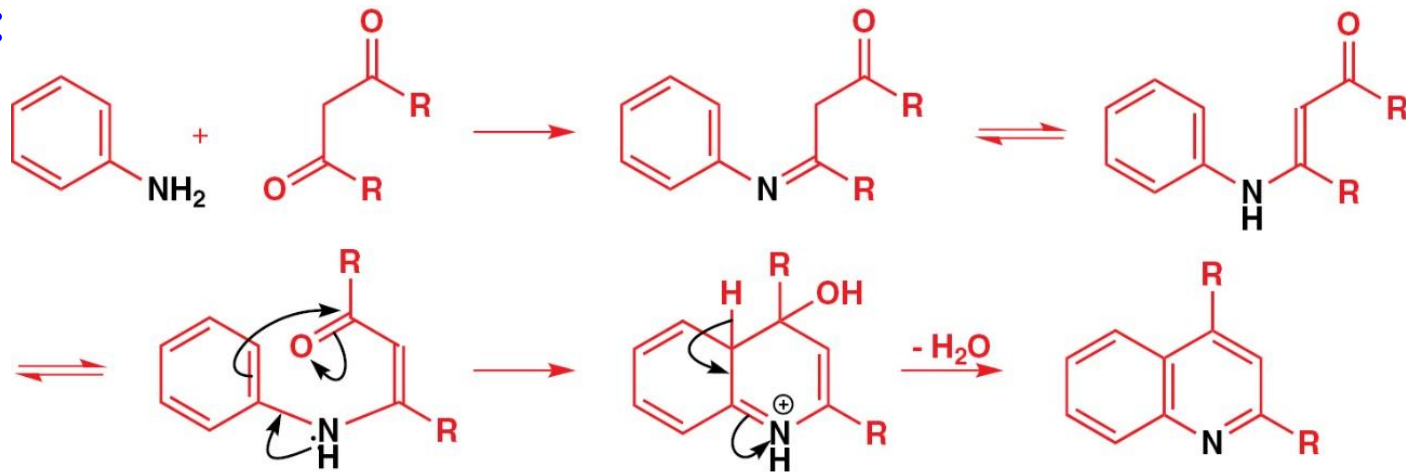


治疗偏头痛药物

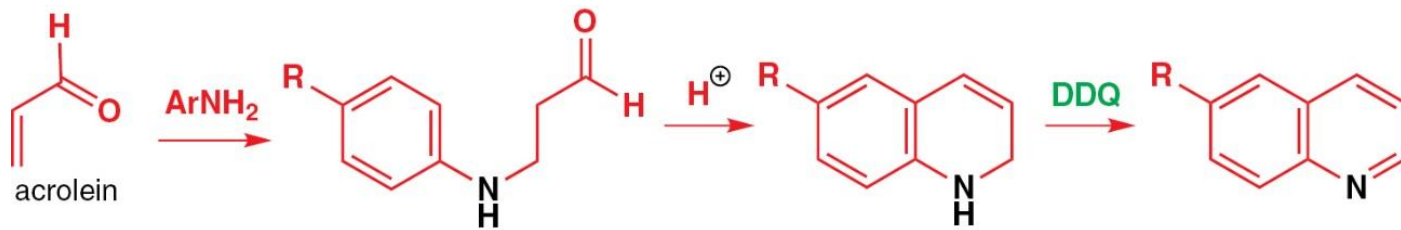


Synthesis of quinolines

方法一:

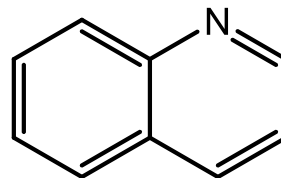
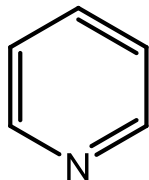


方法二:





7.8 Summary



反应性、选择性、合成

